澳門理工大學
Universidade Politécnica de Macau
Macao Polytechnic University

應用科學學院
Faculdade de Ciências Aplicadas
Faculty of Applied Sciences

Doctor of Philosophy in Computer Applied Technology

Thesis Defence Examination

Thesis Review Comments

Student

Name	Keyan Jin	*ID*	P2317001
Supervisors	Dr. Yapeng Wang		
	Prof. Hugo Gonçalo Oliveira		
Title of Thesis	Improving the Reliability and Accuracy of Dialogue-Based Natural Language Generation		
Date of Submission	2026-05-22		

Reviewer

Name	Dr. Patrick Cheong Iao Pang	*Date*	2026-05-20

Recommendation

☒ Accepted ☐ Suggestion for Revision*

* "Suggestion for Revision" will result in the delay of the oral defence.

Comments

Overview.

While pretrained language models have significantly improved natural language generation fluency, dialogue-based generation remains uniquely challenging because meaning is scattered across multiple speakers, topics overlap, and information relevance shifts dynamically. To address these difficulties, this thesis introduces a sequence of frameworks, including Role-Aware Adapters (RAA), HiSum, and DABART, for improving dialogue-based natural language generation. Notably, the study also reveals that reasoning-oriented large language models do not hold a reliable advantage for dialogue summarization, often producing longer, less concise, and poorly grounded summaries compared to non-reasoning models. Ultimately, the findings highlight that advancement in dialogue-based generation hinges on aligning model behavior with specific task demands, such as content selection, coherence, and structural compression, rather than relying solely on raw generative capacity. Overall, this is a good piece of research.

Knowledge Base and Literature Review.

This body of work systematically addresses the core challenges of dialogue-based generation. Furthermore, a comprehensive evaluation comparing reasoning and non-reasoning large language models reveals empirical insights.

Originality and Independence.

This research demonstrates a high degree of intellectual independence and originality by breaking away from the prevailing paradigm of relying solely on scaled-up, raw generative capacity to solve complex natural language processing tasks. In addition, the work introduces a highly specialized, task-aligned sequence of proprietary methodologies tailored specifically to the structural nuances of multi-turn dialogue.

Arguments and Reasoning.

The thesis is well written in terms of sound arguments and rigorous reasoning, whereby every methodological choice is clearly justified, the limitations of the proposed approaches are appropriately acknowledged.

Evaluation within the Field.

The work has been exceptionally well evaluated within the field, as evidenced by the fruitful publications in high-impact venues, the widespread adoption of the open-source code by the academic community, and the strong endorsement reflected in peer recognition and citations.

Completeness and Coherence.

The thesis exhibits good completeness and coherence, presenting a unified narrative that seamlessly weaves together the literature review, algorithmic developments, and experimental validations into a cohesive and logically flowing body of work.

Conclusion.

This thesis is well-prepared for defence. However, a few minor issues need addressing:

1. Table 8.6: the table caption should explain the meanings of different colour codings.

Additional Comments.